DATASET EXPLORATION

Veda Technology – AI & ML Track | Day 17 – Task 17

Submitted by: Ayush Rajput

Project Topic: Exploratory analysis of the Iris Dataset using Pandas

1. Introduction

Dataset exploration is an important step before machine learning because it helps understand the size, structure, data types, missing values, and statistical characteristics of the data.

2. Objective

To explore the Iris dataset using Pandas functions such as shape, columns, info(), describe(), head(), and tail(), and record meaningful observations.

3. Tools and Technologies

Python, Pandas, Jupyter Notebook, Matplotlib, and the Iris dataset.

4. Dataset Description

The Iris dataset contains measurements of iris flowers. It has 150 rows and 5 columns: sepal length, sepal width, petal length, petal width, and species.

5. Methodology

The dataset was loaded with Pandas, its dimensions and column names were inspected, numerical and categorical columns were identified, df.info() was used for structure and non-null information, df.describe() was used for numerical statistics, and head()/tail() were used to inspect sample records.

6. Dataset Dimensions

Property	Value
Rows	150
Columns	5
Numerical columns	4
Categorical columns	1
Species	3
Missing values	None

7. Observations

1. The dataset has 150 observations and 5 columns.
2. Four features are numerical measurements.
3. Species is a categorical target-like column with three classes: setosa, versicolor, and virginica.
4. No missing values are present in the dataset.
5. Petal length and petal width show noticeable differences between species.
6. Descriptive statistics provide the mean, standard deviation, minimum, quartiles, and maximum for numerical features.

8. Conclusion

The exploration shows that the Iris dataset is clean, compact, and well structured. Performing these checks before modeling helps identify data quality issues and understand the variables that may be useful for later machine-learning tasks.

9. Interview Questions

Why explore a dataset before modeling? To understand its structure, detect missing or incorrect values, identify feature types, and understand distributions before selecting preprocessing and modeling techniques.

What does DataFrame.shape return? A tuple containing the number of rows and columns.

What information does info() provide? It provides the index range, column names, non-null counts, data types, and memory usage.